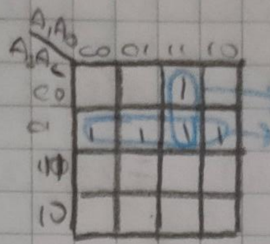
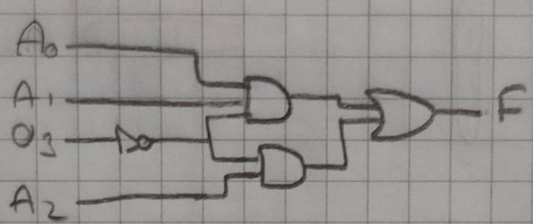
resim 1)

A_3	A_2	A_1	A_0	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0

→ karno H.



→ devre şeması

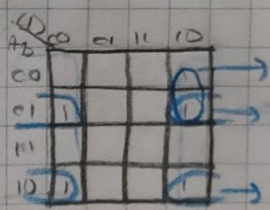


resim 2)

a)

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1

b)



$\overline{A}C\overline{D}$

$\overline{A}B\overline{D}$

$A\overline{B}\overline{D}$

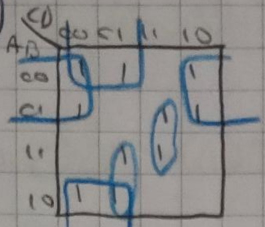
$F = \overline{A}C\overline{D} + \overline{A}B\overline{D} + A\overline{B}\overline{D}$

c)

?

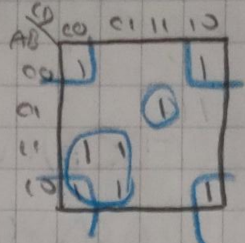
0

1) a-)



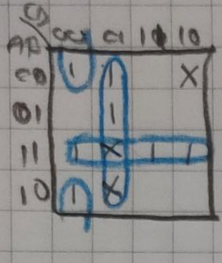
$F = \bar{A}\bar{D} + \bar{B}\bar{C} + BCD + A\bar{C}\bar{D}$

b-)



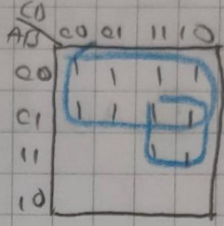
$F\ A\bar{C} + \bar{B}\bar{D} + \bar{A}BCD$

c-)



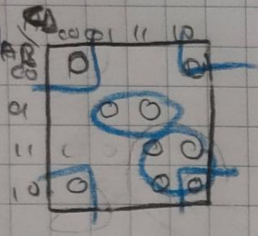
$F = \bar{C}D + AB + \bar{B}\bar{C}\bar{D}$

d-)



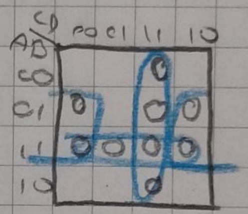
$F = \bar{A} + BC$

2) a-)



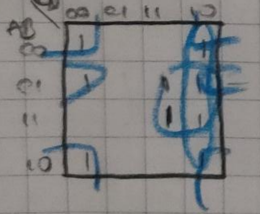
$F = \bar{B}\bar{D} + AC + \bar{A}BD$

b-)



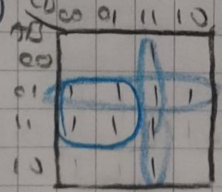
$F = B\bar{D} + AB + CD$

3)

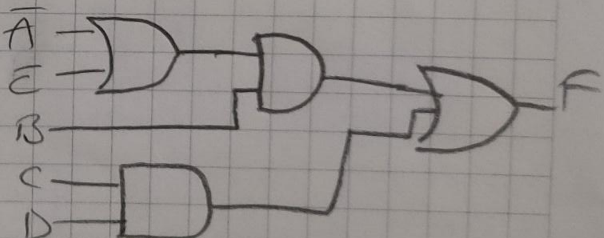


$F = \bar{B}\bar{D} + BC + \bar{A}B\bar{D}$

4)



$F = B\bar{C} + \bar{A}B + CD$

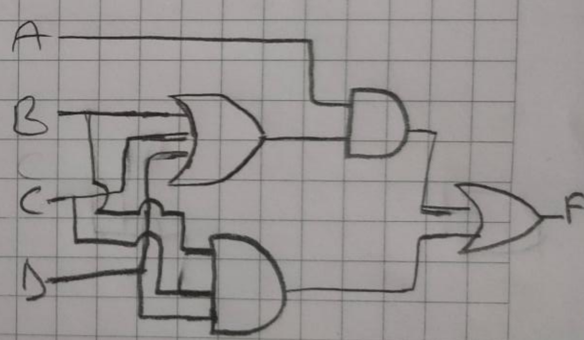


bir banka şubesinde kasanın açılma şartı müdürün yanında bir kişi olması yeterlidir müdür yoksa yardımcısı 2 memur ile kasayı aça bilir memurlar kendi başlarına veya her ikisi birden kasayı açamaz

A → müdür B → müdür yar. C memur 1 d → memur 2 k → kasa

A	B	C	D	k
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

AB \ CD	00	01	11	10
00				
01			1	
11	1	1	1	1
10		1	1	1



$K = AB + AD + AC + BCD$

$K = A \times (B + D + C) + BCD$

$K(A,B,C,D) = \Sigma(7, 9, 10, 11, 12, 13, 14, 15)$

$K(A,B,C,D) = \pi(0, 1, 2, 3, 4, 5, 6, 8)$

11 a) $F = (A+B+C+D)(A+B+C+\overline{D})(A+B+\overline{C}+D)(\overline{A}+B+C+D)(\overline{A}+B+\overline{C}+D$

b) $(\overline{A}\overline{B}CD) + (\overline{A}B\overline{C}\overline{D}) + (\overline{A}B\overline{C}D) + (\overline{A}BC\overline{D}) + (\overline{A}BCD) + (A\overline{B}\overline{C}D) + (A\overline{B}CD)$
$(AB\overline{C}\overline{D}) + (AB\overline{C}D) + (ABC\overline{D}) + (ABCD)$

12 a) $F = (B\overline{C}\overline{D}) + (ABC)$

b) $F = A\overline{B} + AD + \overline{B}D$

13

A \ BC	00	01	11	10
0				1
1	1	1	1	1